Econ 814 Recitation 2 Notes

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1 June 2023 Stochastic Neoclassical Growth model

Consider a stochastic economy that features a representative agent and endogenous labor supply. In this economy, households assign value to both leisure and consumption. The life-time utility function is given by:

$$\sum_{t=0}^{\infty} \beta^t u(c_t, 1 - l_t)$$

The social planner chooses stochastic consumption, labor and capital allocations c_t, l_t, k_{t+1} to maximize life-time utility. Capital depreciates at a constant rate $\delta > 0$. The resource constraint is:

$$c_t + k_{t+1} = (1 - \delta)k_t + z_t k_t^{\alpha} l_t^{1-\alpha}$$

with $\beta \in (0, 1)$ and $\alpha \in (0, 1)$. The technology shock, z_t, follows a N-state Markov chain. Let $Z = z_1, z_2, ...z_N$ be the state space of the Markov chain, and let $P(z_{t+1}|z_t)$ denote the Markov transition matrix of the chain.

1.1 Part A

Formulate the problem of the social planner recursively. State clearly what the state variables and control variables are.

State variables for the social planner: k, z
Control variables: c, l, k'
The recursive formulation is

$$V(k, z) = \max_{c,l,k'} u(c, 1 - l) + \beta V(k', z')$$

with constraint

$$c + k' = (1 - \delta)k + zk^{\alpha} l^{1-\alpha}$$

1.2 Part B

Use the first order conditions and the envelope conditions to derive the Euler equation and the intratemporal optimality condition (in recursive form).

$$u((1-\delta)k + zk^{\alpha}l^{1-\alpha} - k') + \beta V(k', z')$$

Derive w.r.t k'

$$-u_c(\cdot) + \beta V_k(z', k') = 0$$

Derive w.r.t l

$$zk^{\alpha}l^{-\alpha}u_c(\cdot) - u_l(\cdot) = 0$$

Now we want to substitute for $V_k(z', k')$

Derive Bellman w.r.t k

$$V_k(k, z) = (1 - \delta + \alpha z k^{\alpha-1} l^{1-\alpha}) u_c(\cdot)$$

Move forward one period

$$V_k(k', z') = (1 - \delta + \alpha z' k'^{\alpha-1}) l'^{1-\alpha} u_c(\cdot')$$

Plug this into the FOC for k' to get the Euler Equation

$$u_c(c, 1-l) = \beta(1 - \delta + \alpha z' k'^{\alpha-1} l'^{1-\alpha}) u_c(c', 1-l')$$

Paired with the intratemporal labor consumption condition:

$$u_c(c, 1-l) z (k/l)^{\alpha} = u_l(c, 1-l)$$

1.3 Part C

Define a recursive competitive equilibrium, assuming that the firm rents capital from the household at the rental rate, r_t and the household supplies labor at the wage rate, w_t. State the household's and firm's problem. Be sure to clearly identify what are the state and control variables of the household.

Individual state variable for the HH: k
Aggregate state variables: K, Z
Control variables: c, l, k'

The recursive formulation for the HH

$$V(k, K, Z) = \max_{c,l,k'} u(c, 1-l) + \beta V(k', K', Z')$$

subject to the constraint

$$c + k' = (1 - \delta)k + r(K, Z)k + w(K, Z)l$$

and aggregate law of motion $K' = H(K, Z)$

A RCE is the value function $V(k, K, Z)$, policy functions $k'(k, K, Z)$, $l(k, K, Z)$, $c(k, K, Z)$, pricing functions $w(K, Z)$,$r(K, Z)$ and aggregate law of motion such that

$V(k, K, Z)$ solves the HH problem with policy functions above, the pricing functions are the given by the firms problem

$$w(K, Z) = F_l(K, l(k, K, Z))$$
$$r(K, Z) = F_k(K, l(k, K, Z))$$

and markets clear

1.4 Part D

Now assume that households only value consumption and that labor supply is exogenously given by: $l_t = 1$. Also assume that $u(c_t) = \log(c_t)$, $\delta = 1$, and $\log(z') = \lambda_0 + \lambda_1 \log(z) + \epsilon$, ϵ $N(0, \sigma^2)$. Solve the social planner's problem. Use the guess-and-verify method to solve the Bellman equation and derive the associated optimal decision rules for capital accumulation and consumption [Hint: Guess that the value function is equal to a constant plus a logarithmic function in each state variable.]

Take the Bellman equation to be

$$V(k, z) = \log(zk^\alpha - k') + \beta(A + B \log k' + C \log z')$$

Derive w.r.t k'

$$\frac{-1}{zk^\alpha - k'} + \beta B/k' = 0$$

Solve for k' to be

$$k' = \frac{\beta B z k^\alpha}{1 - \beta B}$$

Plug back in to Bellman

$$V(k, z) = \log(zk^\alpha - \frac{\beta B z k^\alpha}{1 - \beta B}) + \beta(A + B \log \frac{\beta B z k^\alpha}{1 - \beta B} + C \log z')$$

Expand and match up terms A B C

$$V(k, z) = 1 \log z + \alpha \log k + \log(1 - \frac{\beta B}{1 - \beta B}) + \beta A + \beta B \log z + \alpha\beta B \log k + \beta B \log(\frac{\beta B}{1 - \beta B}) + C\lambda_1 \log z + C\lambda_0$$

Find that

$$B = \frac{\alpha}{1 - \alpha\beta}$$

$$C = \frac{1 + \frac{\beta\alpha}{1-\alpha\beta}}{1 - \lambda_1}$$

And A is the rest of the constants

1.5 Part E

Describe a computational approach and an algorithm for how to numerically solve the social planner's problem in (d) on a computer.

- Choose an initial guess V_0
- Set iteration counter $j = 0$ and tolerance $\epsilon > 0$
- Main step: on iteration j given values of V_j update V_{j+1}

$$V_{j+1}(k) = \max_{c,k'} u(c) + \beta V_j(k')$$

$$\text{s.t. } c + k' = f(k) + (1 - \delta)k$$

- Continue iterations on the main step until $|V_{j+1} - V_j| < \epsilon$

1.6 Part F

How can we evaluate the performance of our numerical solution in (e) compared to the exact solution obtained in (d)?

Can compare through plotting for through errors in Euler equations

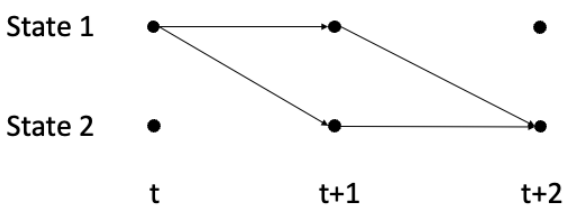
1.7 Part G

Now assume instead that the process for Z is a Markov chain with 2 possible states, $Z = e_1, e_2$, and the following transition matrix:

$$\begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

Compute the following

$$Prob(e_{t+2} = e_2 | e_t = e_1)$$



Where P_{ij} is the probability from moving from point i to point j

$$Prob(e_{t+2} = e_2 | e_t = e_1) = 0.8 * 0.2 + 0.8 * 0.6 = 0.64$$

Define and solve for the stationary distribution π of the Markov chain. Stationary if $\pi_{t+1} = \pi_t$. This implies $\pi = \pi P$ and so $\pi(I - P) = 0$

$$0.8\pi_1 + 0.4\pi_2 = \pi_1$$

$$0.2\pi_1 + 0.6\pi_2 = \pi_2$$

Find they are colinear so add a normalizing $\pi_1 + \pi_2 = 1$ and $\pi_1 = 2/3$, $\pi_2 = 1/3$

2 Hints on next Problem Set

For question 1 there is this alternating preferences in high and low periods. Need to set up 2 value functions:

$$v^H(k_t, z_t) = \max_{c_t, k_t+1} \theta^H \log c_t + \beta v^L(k_{t+1}, z_{t+1})$$

and

$$v^L(k_t, z_t) = \max_{c_t, k_t+1} \theta^L \log c_t + \beta v^H(k_{t+1}, z_{t+1})$$

Plug in the constraint

$$v^i(k_t, z_t) = \max_{k_{t+1}} \theta^i \log(z_t k_t - k_{t+1}) + \beta v^{-i}(k_{t+1}, z_{t+1})$$